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Trends in selective abortions of girls in India: analysis of nationally representative birth histories from 1990 to 2005 and census data from 1991 to 2011

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Summary

Background

India's 2011 census revealed a growing imbalance between the numbers of girls and boys aged 0–6 years, which we postulate is due to increased prenatal sex determination with subsequent selective abortion of female fetuses. We aimed to establish the trends in sex ratio by birth order from 1990 to 2005 with three nationally representative surveys and to quantify the totals of selective abortions of girls with census cohort data.

Methods

We assessed sex ratios by birth order in 0·25 million births in three rounds of the nationally representative National Family Health Survey covering the period from 1990 to 2005. We estimated totals of selective abortion of girls by assessing the birth cohorts of children aged 0–6 years in the 1991, 2001, and 2011 censuses. Our main statistic was the conditional sex ratio of second-order births after a firstborn girl and we used 3-year rolling weighted averages to test for trends, with differences between trends compared by linear regression.

The conditional sex ratio for second-order births when the firstborn was a girl fell from 906 per 1000 boys (99% CI 798–1013) in 1990 to 836 (733–939) in 2005; an annual decline of 0·52% (p for trend=0·002). Declines were much greater in mothers with 10 or more years of education than in mothers with no education, and in wealthier households compared with poorer households. By contrast, we did not detect any significant declines in the sex ratio for second-order births if the firstborn was a boy, or for firstborns. Between the 2001 and 2011 censuses, more than twice the number of Indian districts (local administrative areas) showed declines in the child sex ratio as districts with no change or increases. After adjusting for excess mortality rates in girls, our estimates of number of selective abortions of girls rose from 0–2·0 million in the 1980s, to 1·2–4·1 million in the 1990s, and to 3·1–6·0 million in the 2000s. Each 1% decline in child sex ratio at ages 0–6 years implied 1·2–3·6 million more selective abortions of girls. Selective abortions of girls totalled about 4·2–12·1 million from 1980–2010, with a greater rate of increase in the 1990s than in the 2000s.

Interpretation

Selective abortion of girls, especially for pregnancies after a firstborn girl, has increased substantially in India. Most of India's population now live in states where selective abortion of girls is common.

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